

# XIWEN CHEN

## EDUCATION

xiwenc@usc.edu | GitHub | LinkedIn | Personal Website | Open to Relocate

### University of Southern California

M.S. in Computer Science, Artificial Intelligence Track; GPA: 3.71 / 4.00

Los Angeles, CA

Jan. 2024 – May 2026

### South China Normal University

B.E. in Software Engineering; GPA: 3.87 / 4.00

Guangzhou, China

Sep. 2019 – Jul. 2023

## EXPERIENCE

### Lab for ML, Health and Biomedicine, USC

Los Angeles, CA

Research Lead | Agentic AI, Optimization & Evaluation | Advisor: Prof. Ruishan Liu

May. 2026 – Present

- Lead the development of **RadOncGym**, a **Gymnasium-based agentic optimization environment** for sequential radiotherapy planning, with modular action/state interfaces, **budget-aware control**, and multi-fidelity execution workflows.
- Built a unified **evaluation and experimentation framework** for LLM/RL agents and numerical optimizers, integrating standardized interfaces, automated experiment orchestration, reproducible trajectory logging, and end-to-end model evaluation.
- Extended a **Gurobi-backed** optimization stack into a multi-fidelity simulator with **12 continuous decision variables**; agents iteratively act on surrogate feedback and autonomously decide when to request an expensive full optimization or stop.
- Implemented and evaluated **9 agent configurations** across **3 memory mechanisms** × **3 role structures**, analyzing trajectories and failure modes to quantify how memory/context design and planner/evaluator/critic architectures affect decision quality.
- Benchmarked **3 frontier LLMs** and **1 locally deployed open-source LLM** against black-box optimization policies across **600 contexts**, reducing full evaluations by **81%** while retaining **95%** of downstream objective gains under matched compute budgets.

### 2 Cube Global

Los Angeles, CA

AI Engineer Intern | Speech AI & ML Data Pipelines

Jan. 2026 – May. 2026

- Prototyped and evaluated production-oriented TTS pipelines for multilingual supply-chain notifications, adapting F5-TTS on curated domain data and benchmarking against zero-shot Qwen3-TTS on quality–cost trade-offs to guide model selection.
- Built Python/Librosa speech-data pipelines to transform **80+ hours** of paired audio and transcripts into **45K** aligned utterances, filtering **16%** misaligned samples to create the domain corpus for F5-TTS adaptation and held-out evaluation.

### University of Southern California

Los Angeles, CA

Research Assistant | Data-Centric AI & Training Data Curation | Advisor: Prof. Ruishan Liu

Sep. 2024 – May 2026

- Led two projects on **data-efficient cross-domain learning**, developing **TAVO** and **TAC** to curate high-value source and source–target subsets from **heterogeneous data pools** under label scarcity, constrained training budgets, and distribution shift.
- Designed complementary **budget-aware data-curation** frameworks combining target-conditioned and source-structural signals, validation-guided multi-fidelity search, and active acquisition for compact, target-aligned subset construction.
- Built reproducible **PyTorch/CUDA** training and evaluation pipelines on **BraTS 2021**, **MAMA-MIA**, and **OfficeHome**, supporting domain-shift evaluation, multi-budget ablations, and multi-backbone benchmarking across EfficientViT, nnU-Net, and ResNet-50.
- Across held-out targets at a **12%** source-data budget, TAVO outperformed fixed-selector and domain-adaptation baselines by **+1.9/+2.2 Dice points**, while TAC exceeded the strongest baseline and full-source training by **+1.5/+4.1 points**, respectively.

### Xiaomi Technology Co., Ltd.

Beijing & Shenzhen, China

Research Engineer Intern | Distributed Systems & Cloud Infrastructure

May. 2024 – Sep. 2024

- Integrated network- and erasure-coding components into the enterprise NCDSS **distributed-storage system**, deploying database, coordination, coding, and file-system services and validating K=4,M=2 erasure-coded storage across a **60-node** Linux environment.
- Built a layered reliability and failure-analysis workflow with **50+ client-I/O correctness scenarios** covering encoding, decoding, repair, update, end-to-end file I/O, and recovery under simulated storage failures across **2KB–1GB workloads**.
- Diagnosed system bottlenecks and benchmarked distributed coding implementations across **48 paired tests** and **8 baselines**, improving encoding throughput by **81.5%**, end-to-end upload throughput by **22%**, and recovery latency by **12%**.

## TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Bash, JavaScript

Systems & Infrastructure: Linux, Docker, Git, CUDA, Slurm, Spark, REST APIs, Distributed Systems, Performance Benchmarking

AI/ML & Evaluation: PyTorch, Hugging Face Transformers, scikit-learn, Gymnasium, SGLang, LangGraph, RAG, RLHF, Gurobi

## SELECTED PUBLICATIONS & MANUSCRIPTS

### RadOncGym: An Agent-Based Framework for Radiotherapy Dose-Mimicking Optimization

WACV 2027

First Author | Agentic AI, LLM Agents, Planning & Evaluation, Multi-Fidelity Optimization

Under Review

### TAVO: Target-Aware Source Curation for Budgeted Cross-Center Tumor Segmentation

AAAI 2027

First Author | Data Curation, Data-Efficient Learning, Domain Adaptation, Optimization

Under Review

### Target-Anchored Coverage for Source–Target Curation in Cross-Center Medical Image Segmentation

NeurIPS 2026

First Author | Active Learning, Data Curation, Data-Efficient Learning, Domain Adaptation

Under Review

### A Stroke Image Recognition Model Based on 3D ResNet and Attention Mechanism

JIFS 2022

Second Author | Deep Learning, 3D CNNs, Attention Mechanisms, Image Classification, DOI: 10.3233/JIFS-212511